

Alkanes

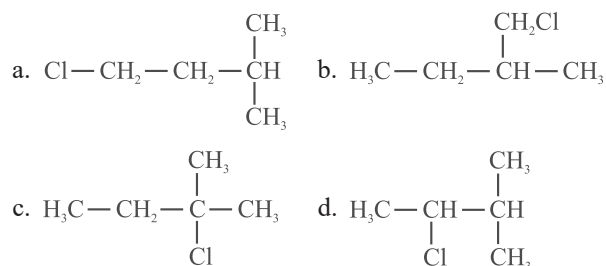
- Dihedral angle of least stable conformer of ethane is: (2021)
 - 180°
 - 60°
 - 0°
 - 120°
- Which of the following alkane cannot be made in good yield by Wurtz reaction? (2020)
 - 2, 3-Dimethylbutane
 - n*-Heptane
 - n*-Butane
 - n*-Hexane
- Which of the following is a free radical substitution reaction? (2020-Covid)
 - Acetylene with HBr
 - Methane with $\text{Br}_2/h\nu$
 - Propene with $\text{HBr}/(\text{C}_6\text{H}_5\text{COO})_2$
 - Benzene with $\text{Br}_2/\text{AlCl}_3$
- Hydrocarbon (A) reacts with bromine by substitution to form an alkyl bromide which by Wurtz reaction is converted to gaseous hydrocarbon containing less than four carbon atoms. (A) is (2018)
 - $\text{CH} \equiv \text{CH}$
 - $\text{CH}_2 = \text{CH}_2$
 - CH_4
 - $\text{CH}_3 - \text{CH}_3$
- With respect to the conformers of ethane, which of the following statements is true? (2017-Delhi)
 - Both bond angles and bond length remains same
 - Bond angle remains same but bond length changes
 - Bond angle changes but bond length remains same
 - Both bond angle and bond length change
- The correct statement regarding the comparison of staggered and eclipsed conformations of ethane, is: (2016 - I)
 - The staggered conformation of ethane is more stable than eclipsed conformation, because staggered conformation has no torsional strain.
 - The staggered conformation of ethane is less stable than eclipsed conformation, because staggered conformation has torsional strain.
 - The eclipsed conformation of ethane is more stable than staggered conformation, because eclipsed conformation has no torsional strain.
 - The eclipsed conformation of ethane is more stable than staggered conformation even though the eclipsed conformation has torsional strain.
- The number of structural isomers possible from the molecular formula $\text{C}_3\text{H}_9\text{N}$ is: (2015 Re)
 - 3
 - 4
 - 5
 - 2

Alkenes

- Compound X on reaction with O_3 followed by $\text{Zn}/\text{H}_2\text{O}$ gives formaldehyde and 2-methyl propanal as products. The compound X is: (2022)
 - Pent-2-ene
 - 3-Methylbut-1-ene
 - 2-Methylbut-1-ene
 - 2-Methylbut-2-ene
- The major product of the following chemical reaction is: (2021)

$$\begin{array}{c} \text{CH}_3 \\ | \\ \text{CH} - \text{CH} = \text{CH}_2 \\ | \\ \text{CH}_3 \end{array} + \text{HBr} \xrightarrow{(\text{C}_6\text{H}_5\text{CO})_2\text{O}_2} ?$$
 - $\begin{array}{c} \text{CH}_3 \\ | \\ \text{CH} - \text{CH}_2 - \text{CH}_2 - \text{O} - \text{COC}_6\text{H}_5 \\ | \\ \text{CH}_3 \end{array}$
 - $\begin{array}{c} \text{CH}_3 \\ | \\ \text{CH} - \text{CH} - \text{CH}_3 \\ | \quad | \\ \text{CH}_3 \quad \text{Br} \end{array}$
 - $\begin{array}{c} \text{CH}_3 \\ | \\ \text{CBr} - \text{CH}_2 - \text{CH}_3 \\ | \\ \text{CH}_3 \end{array}$
 - $\begin{array}{c} \text{CH}_3 \\ | \\ \text{CH} - \text{CH}_2 - \text{CH}_2 - \text{Br} \\ | \\ \text{CH}_3 \end{array}$
- An alkene on ozonolysis gives methanal as one of the product. Its structure is : (2020)
 - $\begin{array}{c} \text{CH}_2 - \text{CH}_2 - \text{CH}_3 \\ | \\ \text{Cyclohexene} \end{array}$
 - $\begin{array}{c} \text{CH}_2 - \text{CH} = \text{CH}_2 \\ | \\ \text{Cyclohexene} \end{array}$
 - $\begin{array}{c} \text{CH}_2\text{CH}_2\text{CH}_3 \\ | \\ \text{Cyclohexene} \end{array}$
 - $\begin{array}{c} \text{CH} = \text{CH} - \text{CH}_3 \\ | \\ \text{Cyclohexene} \end{array}$

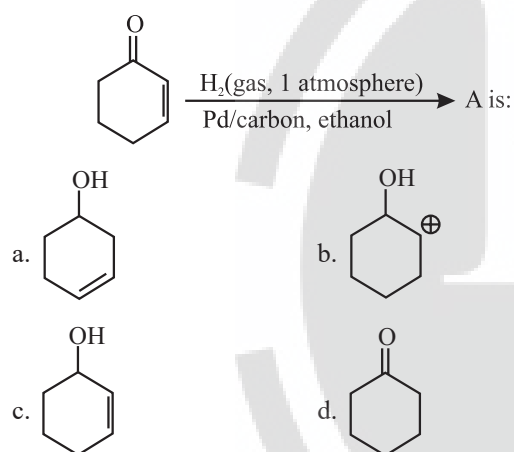
11. An alkene "A" on reaction with O_3 and $Zn - H_2O$ gives propanone and ethanal in equimolar ratio. Addition of HCl to alkene "A" gives "B" as the major product. The structure of product "B" is: (2019)



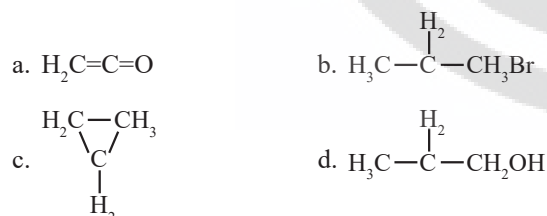
12. The compound that will react most readily with gaseous bromine has the formula: (2016 - II)



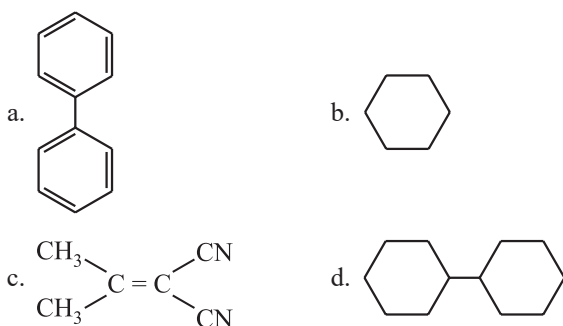
13. The correct structure of the product A formed in the reaction: (2016 - II)



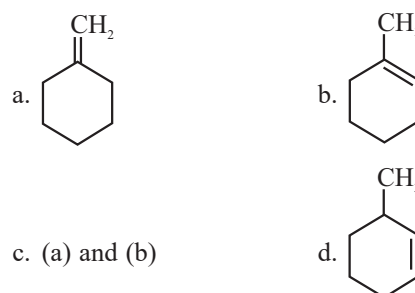
14. Which of the following compounds shall not produce propene by reaction with HBr followed by elimination or direct only elimination reaction? (2016 - II)



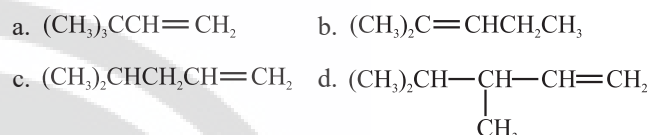
15. In which of the following molecules, all atoms are coplanar? (2016 - II)



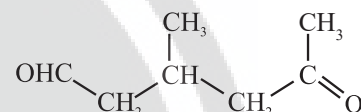
16. In the reaction with HCl, an alkene reacts in accordance with the Markovnikov's rule, to give a product 1-chloro-1-methylcyclohexane. The possible alkene is: (2015 Re)



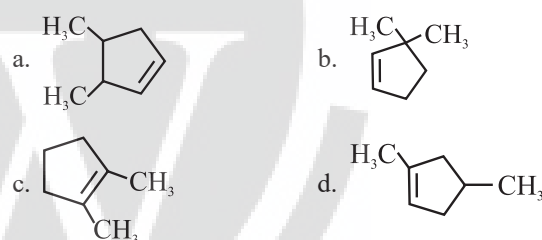
17. 2, 3-Dimethyl-2-butene can be prepared by heating which of the following compounds with a strong acid? (2015 Re)



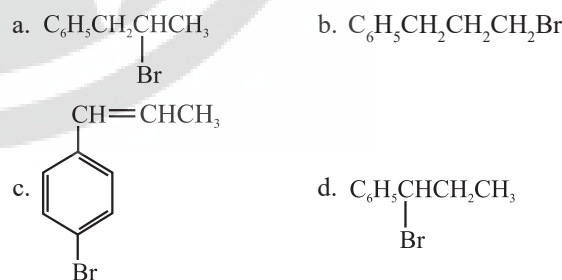
18. A single compound of the structure



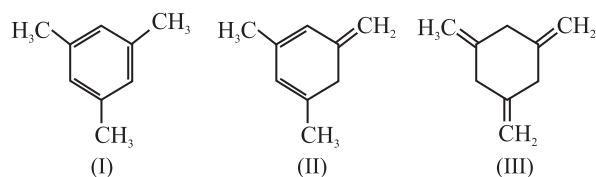
is obtainable from ozonolysis of which of the following cyclic compounds? (2015)



19. The reaction of $C_6H_5CH=CHCH_3$ with HBr produces: (2015)



20. Given:

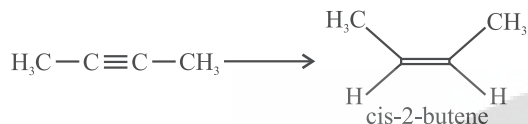


The enthalpy of hydrogenation of these compounds will be in the order as: (2015)

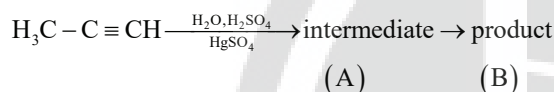
- a. $III > II > I$ b. $II > III > I$
c. $II > I > III$ d. $I > II > III$

Alkynes

21. The number of sigma (s) and pi (p) bonds in pent-2-en-4-yne is
(2019)
- 10 s bonds and 3p bonds
 - 8 s bonds and 5p bonds
 - 11 s bonds and 2p bonds
 - 13 s bonds and no p bonds
22. The most suitable reagent for the following conversion, is:
(2019)

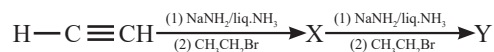


- Na/liquid NH_3
 - H_2 , Pd/C, quinoline
 - Zn/HCl
 - $\text{Hg}^{2+}/\text{H}^+$, H_2O
23. Predict the correct intermediate and product in the following reaction
(2017-Delhi)



- A: $\text{H}_3\text{C}-\underset{\text{SO}_4}{\text{C}}=\text{CH}_2$ B: $\text{H}_3\text{C}-\underset{\text{O}}{\text{C}}-\text{CH}_3$
- A: $\text{H}_3\text{C}-\underset{\text{OH}}{\text{C}}=\text{CH}_2$ B: $\text{H}_3\text{C}-\underset{\text{O}}{\text{C}}-\text{CH}_3$
- A: $\text{H}_3\text{C}-\underset{\text{OH}}{\text{C}}=\text{CH}_2$ B: $\text{H}_3\text{C}-\underset{\text{SO}_4}{\text{C}}=\text{CH}_2$
- A: $\text{H}_3\text{C}-\underset{\text{O}}{\text{C}}-\text{CH}_3$ B: $\text{H}_3\text{C}-\text{C}\equiv\text{CH}$

24. Which one is the correct order of acidity? (2017-Delhi)
- $\text{CH}_3-\text{CH}_3 > \text{CH}_2=\text{CH}_2 > \text{CH}_3-\text{C}\equiv\text{CH} > \text{CH}\equiv\text{CH}$
 - $\text{CH}_2=\text{CH}_2 > \text{CH}_3-\text{CH}=\text{CH}_2 > \text{CH}_3-\text{C}\equiv\text{CH} > \text{CH}\equiv\text{CH}$
 - $\text{CH}\equiv\text{CH} > \text{CH}_3-\text{C}\equiv\text{CH} > \text{CH}_2=\text{CH}_2 > \text{CH}_3-\text{CH}_3$
 - $\text{CH}\equiv\text{CH} > \text{CH}_2=\text{CH}_2 > \text{CH}_3-\text{C}\equiv\text{CH} > \text{CH}_3-\text{CH}_3$
25. In the reaction: (2016 - I)

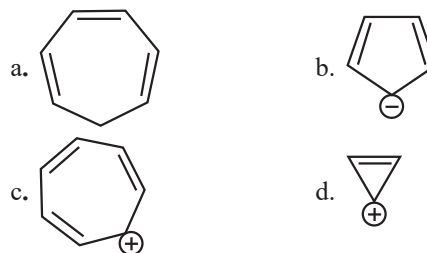


- X = 1-Butyne ; Y = 2-Hexyne
 - X = 1-Butyne ; Y = 3-Hexyne
 - X = 2-Butyne ; Y = 3-Hexyne
 - X = 2-Butyne ; Y = 2-Hexyne
26. The pair of electron in the given carbanion, $\text{CH}_3\text{C}\equiv\text{C}^-$, is present in which of the following orbitals? (2016 - I)

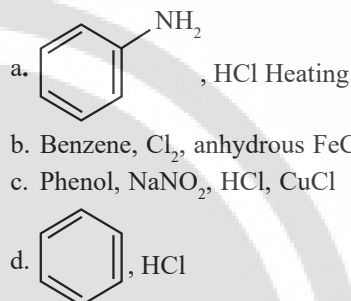
- sp
- d^2sp
- sp^3
- sp^2

Aromatic Hydrocarbons

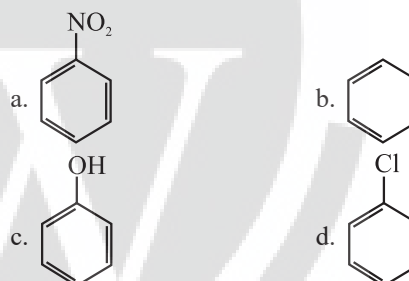
27. Which compound amongst the following is not an aromatic compound? (2022)



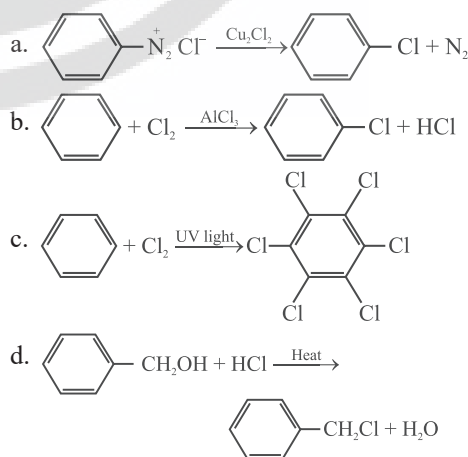
28. Which of the following is suitable to synthesize chlorobenzene? (2022)



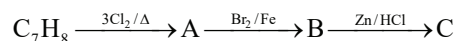
29. Which of the following compound is most reactive in electrophilic aromatic substitution? (2020-Covid)



30. Among the following, the reaction that proceeds through an electrophilic substitution, is: (2019)



31. The compound C_7H_8 undergoes the following reactions: (2018)



The product 'C' is

- 6 p-orbitals and 6 unpaired electrons
- 7 p-orbitals and 6 unpaired electrons
- 7 p-orbitals and 7 unpaired electrons
- 6 p-orbitals and 7 unpaired electrons

[illegible]